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MASTER'S DISSERTATIONS

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# The (Re)centralisation of Health Services in Mexico

*A Difference-in-Differences Analysis of Household Out-of-Pocket Spending*

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*Candidate Number:*  
73197

*Supervisor:*  
Dr. Linda Li

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LONDON SCHOOL OF ECONOMICS AND POLITICAL SCIENCE

## *Abstract*

Department of Methodology

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### **The (Re)centralisation of Health Services in Mexico**

by 73197

Mexico's decision to dismantle Seguro Popular and consolidate health-service administration for the population without contributory social security—first under the Health Institute for Welfare (INSABI) in 2020, and subsequently under an expanded IMSS-Bienestar—was justified in large part as a means of protecting households from the financial burden of paying for care out of pocket. This research tests that premise directly, exploiting the staggered, state-by-state adoption of the reform to compare out-of-pocket health spending in states that transferred administration to the federal government against the eight states that did not. Using household survey data spanning 2016 to 2024 and a difference-in-differences design, it finds that affiliation with the centralised institutions is associated not with a reduction but with a 23.0% increase in expected household out-of-pocket spending, an effect visible across all three spending subcategories—medicines, hospitalisation, and outpatient care—statistically strongest for medicines and largest in absolute pesos for outpatient/ambulatory care. This result, while not yet fully robust to how the reform's ambiguous transition year is treated, sits awkwardly against the government's own stated goal of reducing out-of-pocket exposure by 2030, and speaks directly to a longstanding debate over whether centralising a historically fragmented and decentralised health system relieves the pressures on household finances or, through the erosion of local responsiveness and accountability, ends up reproducing them under a different institutional roof.

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# List of Abbreviations

|                       |                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>CAUSES</b>         | Catálogo Universal de Servicios de Salud (Universal Catalogue of Health Services)                                              |
| <b>CHE</b>            | Catastrophic Health Expenditure                                                                                                |
| <b>CI</b>             | Confidence Interval                                                                                                            |
| <b>CONEVAL</b>        | Consejo Nacional de Evaluación de la Política de Desarrollo Social (National Council for Social Development Policy Evaluation) |
| <b>COVID-19</b>       | Coronavirus Disease 2019                                                                                                       |
| <b>ENIGH</b>          | Encuesta Nacional de Ingresos y Gastos de los Hogares (National Household Income and Expenditure Survey)                       |
| <b>ENSANUT</b>        | Encuesta Nacional de Salud y Nutrición (National Health and Nutrition Survey)                                                  |
| <b>IHE</b>            | Impoverishing Health Expenditure                                                                                               |
| <b>IMSS</b>           | Instituto Mexicano del Seguro Social (Mexican Institute for Social Security)                                                   |
| <b>IMSS-Bienestar</b> | IMSS para el Bienestar (IMSS for Well-being)                                                                                   |
| <b>INEGI</b>          | Instituto Nacional de Estadística y Geografía (National Institute of Statistics and Geography)                                 |
| <b>INSABI</b>         | Instituto de Salud para el Bienestar (Health Institute for Welfare)                                                            |
| <b>ISSSTE</b>         | Instituto de Seguridad y Servicios Sociales de los Trabajadores del Estado (Social Security Institute for State Workers)       |
| <b>OLS</b>            | Ordinary Least Squares                                                                                                         |
| <b>OOP</b>            | Out-of-Pocket                                                                                                                  |
| <b>OOPHE</b>          | Out-of-Pocket Health Expenditure                                                                                               |
| <b>PHS</b>            | Public Health System                                                                                                           |
| <b>PND</b>            | Plan Nacional de Desarrollo (National Development Plan)                                                                        |
| <b>PPML</b>           | Poisson Pseudo-Maximum-Likelihood                                                                                              |
| <b>PROSESA</b>        | Programa Sectorial de Salud (Health Sectoral Programme)                                                                        |
| <b>SE</b>             | Standard Error                                                                                                                 |
| <b>UHC</b>            | Universal Health Coverage                                                                                                      |
| <b>UN</b>             | United Nations                                                                                                                 |
| <b>WLS</b>            | Weighted Least Squares                                                                                                         |



## Chapter 1

# Institutional Change, Fragmentation, and Financial Protection

### 1.1 Introduction

Mexico’s decision to dismantle Seguro Popular and consolidate health-service administration for the population without contributory social security—first under the Health Institute for Welfare (INSABI) in 2020, and subsequently under an expanded IMSS-Bienestar—was justified in large part as a means of protecting households from the financial burden of paying for care out of pocket. This research tests that premise directly, exploiting the state-by-state adoption of the reform to compare out-of-pocket health spending in states that transferred administration to the federal government against the eight states that did not. Using household survey data spanning 2016 to 2024 and a difference-in-differences design, it finds that affiliation with the centralised institutions is associated not with a reduction but with a 23.0% increase in expected household out-of-pocket spending, an effect visible across all three spending subcategories — medicines, hospitalisation, and outpatient care, statistically strongest for medicines and largest in absolute pesos for outpatient care. This result, while not yet fully robust to how the reform’s ambiguous and atypical transition year is treated, sits awkwardly against the government’s own stated goal of achieving Universal Health Coverage (UHC) and reducing out-of-pocket exposure by 2030, and speaks directly to a longstanding debate over whether centralising a historically fragmented and decentralised health system relieves the pressures on household finances by increasing economies of scale and control or, through the disruption of local responsiveness and the loss of state-level accountability, simply relocates the financial burden to households.

Since the transfer of INSABI’s functions to IMSS-Bienestar (IMSS-B) in 2022–2023, current President Claudia Sheinbaum Pardo has continued her predecessor’s health policy, deepening the centralisation and integration of state health systems into IMSS-Bienestar — the branch of the Mexican Social Security Institute now responsible for the population without contributory social security. The stated objectives of this reform are to standardise care protocols, distribute resources more equitably across the territory, modernise infrastructure, and build interoperable health information systems (World Bank 2025). While these reforms promise progress toward universal coverage and improved access, six opposition-governed states have so far declined to transfer the administration of their health services to the federal government, preserving a

degree of institutional heterogeneity that is central to the empirical strategy of this research.

This variation in the timing and extent of states' adherence to the federal reform — some states transferring primary, secondary, and even tertiary care to the centre, others retaining full administrative autonomy — offers a compelling natural experiment through which to study the effects of (re)centralised governance on the delivery of public health services. The remainder of this paper proceeds as follows. Section 1.2 situates the discussion within the broader problem of fragmentation that has historically characterised the Mexican health system. Section 1.3 traces the institutional trajectory from Seguro Popular to INSABI and then to IMSS-Bienestar, and describes the stated policy goals for 2030. Section 1.4 reviews the evidence on out-of-pocket health expenditure, the principal channel through which households feel fragmentation and reform. Section 1.5 reviews the theoretical debate on (de)centralisation of public services, laying the groundwork for the empirical analysis presented in the sections that follow. Chapter 2 describes the research design and the data used to identify the effect of centralisation of health services. Section 3.2 describes the results for our model using Poisson pseudo-maximum-likelihood (PPML) in Out-of-Pocket Health Expenditure (OOPHE). Section 3.4 discusses the findings against the theoretical expectations and policy goals. Finally, Section 3.5 presents the conclusions and final remarks.

## 1.2 Fragmented Health Systems

Fragmentation is widely regarded as the central structural problem of the Mexican health system (Gómez-Dantés et al. 2023). Following Bossert et al. (2014), fragmentation refers to the coexistence of several parallel health-care delivery subsystems that serve different population groups, are financed through separate pools of resources, and operate under distinct rules governing access to benefits and funding. This configuration is not unique to Mexico: most health systems in Latin America share a similar architecture, generally the legacy of health and social-security institutions created between the 1930s and 1950s that combined a well-resourced, contributory social-security branch offering comprehensive care as a labour right, with a Ministry-of-Health branch offering a narrower package of services, historically financed and justified as public charity, to the population outside the formal labour market (Gómez-Dantés et al. 2023).

In Mexico, this translated into a system with, on one side, the contributory social-security institutions — the Mexican Institute for Social Security (IMSS) for private-sector workers, the Institute for Social Security and Services for State Workers (ISSSTE) for civil servants, and smaller institutes for the employees of the Mexican oil company (Pemex), the army (Sedena), and the navy (Semar) — covering approximately 56.6 million people; and, on the other, the federal and state ministries of health together with the historically small IMSS-Bienestar programme for the most deprived rural communities, covering a comparable population lacking contributory social security (Gómez-Dantés et al. 2023). A private sector operating on a fee-for-service or private-insurance basis serves those with the ability to pay, but in Mexico only cover 11% of the population (Asociación Mexicana de Instituciones de Seguros 2026). The institutional arrangement of the social-security branch has remained largely unchanged since the 1940s and 1960s, while the one for covering the non-salaried population has undergone three major transformations in the past two decades: the creation of Seguro Popular

in 2003–2004, its replacement by INSABI in 2019–2020, and the subsequent transfer of INSABI’s responsibilities to an expanded IMSS-Bienestar in 2022–2023.

The consequences of fragmentation are well documented. Because financial resources per capita differ substantially across subsystems — with the social-security branch generally better funded than the services available to the non-salaried population — fragmentation produces persistent gaps in health benefits, quality of care, and financial protection between population groups defined essentially by their labour-market status rather than by health need (Gómez-Dantés et al. 2023). Comparative evidence from twelve Latin American countries shows that the share of households facing catastrophic health expenditure under fragmented arrangements ranges from 1% to 25%, with the burden falling disproportionately on rural, poor, and elderly households, and on those without social security (Knaul et al. 2011). Fragmentation is also associated with inefficiency, since duplicated administrative and clinical infrastructure tends to be under-utilised within each subsystem, and with an erosion of the state’s capacity to monitor performance, hold providers accountable, and articulate a coherent national vision for the sector (Gómez-Dantés et al. 2023; World Bank 2025).

Historically, part of Mexico’s fragmentation problem has been layered onto — and, according to some accounts, worsened by — the decentralisation of health services to the states that began in the 1980s and deepened in the 1990s, which produced 32 state-level health systems of highly uneven administrative and fiscal capacity (World Bank 2025). The 2003 reform that created Seguro Popular preserved this decentralised operational structure: states retained responsibility for administering personnel, supplies, and infrastructure, and financial resources were transferred through federal–state coordination agreements calibrated to the number of affiliated beneficiaries rather than to service quality or health outcomes (World Bank 2025). Whether the current reform — which reverses this decentralisation by consolidating service delivery under a single federal operator — succeeds in reducing fragmentation, or whether it merely relocates the underlying coordination problem from the federal–state axis to the intra-federal one, is an open empirical question that motivates this research.

## 1.3 Health Insurance in Mexico

### 1.3.1 From Seguro Popular to INSABI

The 2003 amendment to the General Health Law established, for the first time, a right for all Mexicans to be incorporated into the National System of Social Protection in Health, and created Seguro Popular as its financing vehicle (Diario Oficial de la Federación 2003). Seguro Popular was a voluntary public insurance scheme targeted at the population without contributory social security. Affiliation required an application, an identification document, and a fee scaled to the household’s ability to pay; enrolled households gained access to first-level medical services and to an expanding package of interventions set out in the Universal Catalogue of Health Services (CAUSES), which grew from 91 covered interventions in 2004 to 255 in 2007 and 294 by 2018 (Knaul et al. 2023). A complementary fund for catastrophic expenditures, created alongside Seguro Popular in 2004, expanded from eight covered conditions to 66 by 2018, and a dedicated children’s health fund (Seguro Médico para una Nueva Generación / 21st Century Medical Insurance) was added in 2007 (Knaul et al. 2023).

Seguro Popular’s growth was not linear, and its trajectory can be divided into distinct phases with different equity and coverage implications. In its early, more focalised years

(2003–2010), the programme prioritised enrolling the poorest households — by 2011, 52.6 million people were affiliated, most drawn from the bottom four income deciles, with disproportionate representation from rural and indigenous communities — and this period is associated with the clearest gains in financial protection and redistribution (Knaul et al. 2012).<sup>1</sup> From around 2010 onward, however, continued expansion pushed enrolment past its originally targeted population, reaching roughly 51 million affiliates by 2019 and, together with contributory social security, covering close to 98% of the population by some estimates (Olvera et al. 2026, citing CONEVAL 2024; Knaul et al. 2012). Garcia-Diaz et al. (2023) show that this later phase (2010–2014) coincided with a weakening, rather than a strengthening, of the programme’s redistributive and vertical-equity effects, a pattern they attribute to oversaturation: as Seguro Popular grew beyond its initially targeted, poorest beneficiaries, its financing and benefit structure struggled to keep pace, diluting the focalised gains achieved in its first decade and producing adverse re-ranking effects among insured families. By the programme’s final years, then, Seguro Popular faced a structural tension between the political and social pressure to expand coverage as close to universally as possible and the financing model’s limited capacity and fiscal space to sustain equitable, quality benefits at that scale (Garcia-Diaz et al. 2023).

Operationally, Seguro Popular was administered through a decentralised scheme: states were responsible for providing medical services through their own health systems and for applying transferred resources, while retaining substantial autonomy over personnel, supplies, and financial administration; the federal government’s role was largely regulatory and normative (Olvera et al. 2026). By the mid-2010s, Seguro Popular — together with the contributory social-security institutions — had reduced the uninsured share of the population to below 10%, reaching a peak enrolment above 50 million affiliates in 2015 and, combined with social security, covering approximately 110.9 million people by 2016 (Knaul et al. 2023; Gómez-Dantés et al. 2023).

In November 2019, the federal government announced the cancellation of Seguro Popular and its replacement by INSABI, formally created through a reform of the General Health Law and launched in January 2020 (DOF 2019). The change entailed three substantive modifications. First, the individual affiliation process was eliminated: since access to INSABI was in principle universal for anyone without contributory social security, no registration or identification requirement remained in place (DOF 2019). Second, the financing mechanism changed: whereas Seguro Popular had been financed through a combination of user fees and state and federal contributions tied to the number of affiliated beneficiaries, INSABI eliminated user fees and replaced affiliate-linked transfers with annual, state-specific coordination agreements between each state government and the federal government (DOF 2019; Cortés-Adame and Gómez-Dantés 2025). Third, and most consequential for the purposes of this research, the reform altered the role of the states in administering the health system: each of Mexico’s 32 states could decide, through its coordination agreement, whether to retain administrative responsibility for primary, secondary, and tertiary care, or to transfer some or all of these levels to the newly created federal institute.

This state-level discretion produced staggered and heterogeneous adoption. Twenty-one states signed coordination agreements transferring authority to INSABI in 2020, five more joined in 2021, and eight states — Aguascalientes, Coahuila, Chihuahua,

<sup>1</sup>Percentages in this paragraph refer to enrolment among the population not covered by contributory social security or private insurance, as estimated from INEGI’s Encuesta Nacional de Ingresos y Gastos de los Hogares (ENIGH); see Knaul et al. (2023), Figure 1.

Durango, Guanajuato, Jalisco, Nuevo León and Querétaro, among the states that retained full administrative control across all three levels of care — opted to retain administration of their health services. Most of the states that transferred authority did so for primary and secondary care, the levels of care on which the extinct Seguro Popular had concentrated its benefits.

The initial results of INSABI's implementation were troubling by most accounts. According to CONEVAL's measurement of multidimensional poverty (Gómez-Dantés et al. 2023),<sup>[^coneval]</sup> the population without access to health services grew from 20.1 million (16%) in 2018 to 35.7 million (28%) in 2020 — an increase of 15.6 million people in two years. Much of this deterioration has been attributed to the coincidence of the reform with the onset of the COVID-19 pandemic, to medicine-procurement failures after purchasing responsibility was shifted first to the Ministry of Finance and then to the UN Office for Project Services, and to the gap between financing and the number of beneficiaries served, which left states with fewer resources to sustain service levels even as demand for care continued (Knaul et al. 2023; CIEP 2021).

### 1.3.2 From INSABI to IMSS-Bienestar

By early 2022, facing continued implementation difficulties, the federal government announced its intention to expand and strengthen IMSS-Bienestar — until then a comparatively small programme that, for nearly forty years, had provided limited primary-care services to around 12 million people in rural and low-income urban areas — so that it could absorb INSABI's responsibilities for the population without contributory social security (Gómez-Dantés et al. 2023; Knaul et al. 2023). The transition was formalised through a presidential decree of August 2022 creating the decentralised public entity Health Services of the Mexican Social Security Institute for Well-being (IMSS-BIENESTAR in Spanish), and INSABI was legally dissolved in 2023, with its functions and remaining coordination agreements transferred to the new entity (DOF 2022).

The transition amounted to a tacit acknowledgement that INSABI had not achieved its stated objectives (Knaul et al. 2023). Before the 2022 reform, IMSS-Bienestar, unlike Seguro Popular, was managed directly by IMSS — the same institution that provides comprehensive care to the salaried population — yet its non-salaried beneficiaries do not gain access to the infrastructure or benefits available to IMSS's contributory affiliates, a structural asymmetry that several authors have described as a form of institutional segregation within a single managing agency (Knaul et al. 2023). At the time of the 2022 transition, IMSS-Bienestar's own historical coverage of its target rural population had fallen — from 83% in 2015 to 67% in 2020 — even as its budget nearly doubled, from roughly 13.6 billion pesos in 2021 to close to 24 billion pesos in 2022, a sum that nonetheless remained well below historical Seguro Popular funding levels once adjusted for the size of the population to be served (Knaul et al. 2023). As of August 2022, IMSS-Bienestar became a non-sectorized decentralised public body operated in only 19 of 32 states and initially lacked the administrative capacity to absorb the human and material resources of the 24 states that had operated under INSABI (Knaul et al. 2023).

Importantly for the empirical design of research exploiting this reform sequence, recent studies of the Mexican (re)centralisation — including this one — focus on the 2019–2020 discontinuity between Seguro Popular and INSABI rather than on the subsequent 2022–2023 transition to IMSS-Bienestar, because the earlier reform preserved the clean cross-state variation in adoption timing described above, whereas the later

transition maintained the same centralised model of service provision across virtually all states that had already joined INSABI (Olvera et al. 2026). The consolidation into IMSS-Bienestar is nonetheless the institutional endpoint against which the present-day system must be understood.

### 1.3.3 Goals for 2030

The stated ambition of the current phase of reform is to achieve Universal Health Coverage in Mexico by 2030 through the progressive integration of all public health institutions serving the non-salaried population into IMSS-Bienestar, framed as a “Public Health System” (PHS) (World Bank 2025). This objective is formally embedded in the National Development Plan (Plan Nacional de Desarrollo, PND) 2025–2030 and the Health Sectoral Programme (Programa Sectorial de Salud, PROSESA) 2025–2030, both of which identify guaranteeing the constitutional right to health protection, ensuring financial protection against illness, and consolidating a modern, integrated, and equitable health system as central objectives of the government’s social agenda (World Bank 2025).

The reform is conceived as a two-phase process. The first phase, launched in 2022, corresponds to the federalisation of service delivery already described in Section 1.3: the transfer of facilities, personnel, and budgets from state health systems to IMSS-Bienestar, intended to establish a uniform framework for the quality and governance of care, standardised clinical protocols, a more equitable geographic distribution of resources, modernised infrastructure, and interoperable health-information systems (World Bank 2025). As of 2025, official figures place overall population coverage — through IMSS, ISSSTE, the expanding IMSS-Bienestar network, and other social-security institutions combined — at approximately 87%, with the remaining 13% of the population, concentrated among informal workers, rural residents, and marginalised communities in the eight states that have not signed integration agreements, continuing to be served through fragmented, state-specific delivery arrangements (World Bank 2025). It is worth flagging, for the sake of internal consistency in this research, that this 87% administrative coverage estimate for 2025 sits in evident tension with household-survey-based estimates — such as those reported by INEGI and discussed in Section 1.1 and Section 1.4 — that point to a considerably lower and, on some measures, declining share of the population reporting effective access to health services over the same period.<sup>2</sup>

Beyond coverage, the government has also identified persistently high out-of-pocket spending as a central obstacle to financial protection and a driver of catastrophic and impoverishing expenditure, and has framed its reduction as a measurable target of the 2030 agenda (World Bank 2025). Mexican government is committed to reduce the percentage of the population that received care in the public sector and had out-of-pocket health expenses from 17.8% in 2022, to 12.0% by 2028, and to 10.0% by 2030. In parallel, the government aims to reduce the percentage of the population without access to health care services from 39.09% in 2022, to 10.0% by 2028, and 5.0% by 2030; among other goals related to improve population health outcomes (World Bank 2025). However, private expenditure is estimated to represent close to 48% of total

<sup>2</sup>Administrative coverage figures (e.g., the ~87% figure reported for 2025 in official government communications) and household-survey-based effective-access figures are not directly comparable and have diverged since 2018. Any single “coverage” figure cited without specifying its source and definition should be treated with caution; this research uses INEGI/ENIGH- and ENSANUT-based measures wherever the underlying outcome is household-reported access or expenditure, and reserves administrative figures for describing stated institutional coverage.

health expenditure in Mexico, of which out-of-pocket payments account for roughly 41 percentage points and private health insurance for approximately 7 percentage points (World Bank 2025). Out-of-pocket spending remains a major barrier to financial protection and is among the highest in OECD countries, pushing families into poverty and discouraging timely care-seeking, with Mexico’s total health expenditure about 6% of GDP—only 3% from public sources—far below the OECD average (World Bank 2025).

## 1.4 Financial protection in health: Out-Of-Pocket Health Expenditure

Out-of-pocket health expenditure (OOPHE) is widely regarded, in both the theoretical and empirical literature on health financing, as the least equitable and least efficient mechanism for financing health care, precisely because it offers no risk pooling and therefore exposes households to expenditure shocks unrelated to their ability to pay (Garcia-Diaz et al. 2023). This burden falls disproportionately on households in low- and middle-income countries, where the absence of adequate risk-pooling mechanisms is closely tied to the segmentation and fragmentation of the health system (Garcia-Diaz et al. 2023). Reducing the share of OOPHE in total health financing, and thereby the incidence of catastrophic and impoverishing health expenditure, is accordingly one of the principal justifications advanced for the social-insurance reforms undertaken across Latin America since the early 2000s, and is explicitly incorporated into the Sustainable Development Goal 3.8.2 indicators of financial risk protection (Garcia-Diaz et al. 2023).

Seguro Popular was designed with exactly this objective in mind, and the empirical literature evaluating its first fifteen years is comparatively consistent in finding that it achieved measurable — if uneven — success. National-level indicators of catastrophic health expenditure (CHE, conventionally defined as health spending exceeding 30% of a household’s capacity to pay) and impoverishing health expenditure (IHE, health spending that pushes a household below, or further below, the poverty line) both trended downward between 1992 and 2010, alongside a narrowing of the gap in financial protection between households with and without social security (Knaul et al. 2012). Using propensity-score matching and quasi-experimental designs, several studies attribute part of this improvement directly to Seguro Popular enrolment, alongside gains in reported utilisation, self-assessed care quality, and — in some studies — reductions in mortality (Garcia-Diaz et al. (2023), summarising King et al., Sosa-Rubí et al., Wirtz et al., and others; Colchero et al. 2022).

More recent equity-focused analyses complicate this picture. Using eleven waves of Mexico’s National Household Income and Expenditure Survey (2002–2020) and propensity-score matching to compare Seguro Popular-affiliated and uninsured households without social security, Garcia-Diaz et al. (2023) find that the redistributive effect of Seguro Popular on out-of-pocket spending was positive and pronounced during the programme’s early, more targeted years (2003–2010), but weakened after 2010, plausibly because rapid growth in enrolment diluted the programme’s initially progressive focus. The same study documents that Seguro Popular was, throughout its existence, more horizontally inequitable than the uninsured comparison group — meaning that households with similar spending capacity received unequal treatment under the scheme — even though it remained progressive in vertical-equity terms in most years. These findings, together with related work by García-Diaz (2022)

on the determinants of effective access and catastrophic expenditure using the 2018 National Health and Nutrition Survey (ENSANUT), suggest that the programme's financial-protection gains, while real, were neither complete nor evenly distributed — a caveat relevant to any normative comparison between Seguro Popular and its successors.

According to most of the reviewed evidence, the transition from Seguro Popular to INSABI coincided with a marked deterioration in financial protection. The share of households facing excessive health spending (the combination of catastrophic and impoverishing expenditure) nearly doubled between 2018 and 2020, from 2.4% to 4.4%, a level not observed since 2004, the first year of Seguro Popular's operation (Knaul et al. 2023). The increase was substantially larger among the uninsured than among households with contributory social security, and the number of households without any health insurance rose from about 3 million (8.8%) in 2019 to 5.8 million (16.5%) in 2020 — the sharpest jump recorded since the early 2000s (Serván-Mori et al. 2023). Serván-Mori et al. (2023) finds statistically significant increases in catastrophic, impoverishing, and excessive health expenditure concentrated almost entirely among households without health insurance, with a pronounced geographic pattern affecting central and southern states — including Mexico state, Mexico City, Morelos, Puebla, Guerrero, Oaxaca, and Chiapas — more severely than the wealthier northern states.

Average out-of-pocket spending among health-service users increased by 75% between the 2018 and 2021 rounds of ENSANUT: spending on outpatient visits rose by 60%, on medicines by 49%, on medical testing by 96%, and on other medical expenses, including hospitalisation, by nearly 120% (Knaul et al. 2023). Waiting times for outpatient services, a proxy for the quality of access rather than its financial cost, also deteriorated after the reform: national average waiting times for the population previously covered by Seguro Popular rose from 70 minutes in 2018 to 84 minutes in 2021, reversing a downward trend observed since 2012 (Knaul et al. 2023). Consistent with these access difficulties, the share of the population reporting a health problem who chose to use private, out-of-pocket services rather than public facilities — already substantial under Seguro Popular, at around 59–60% in 2018 and 2021 — appears to have risen further as confidence in, and availability of, public services covered under INSABI declined (Knaul et al. 2023).

Taken together, this body of evidence establishes two points that are central to the remainder of this research. First, the erosion of financial protection after 2018 was not a general phenomenon affecting the Mexican population uniformly, but one concentrated specifically among households without contributory social security — precisely the population whose health-care provision was reassigned, state by state and at different points in time, between decentralised state administration and the federal government's new institutes. Second, the deterioration exhibits a clear territorial pattern, correlated with pre-existing socioeconomic disadvantage, suggesting that the effects of the 2019–2020 reform were not homogeneous across the states that adopted it — a heterogeneity that Section 1.5 relates to the broader theoretical literature on the conditions under which (re)centralisation improves or worsens public-service delivery.

## 1.5 The (De)centralisation Debate

The classic theory of fiscal federalism provides the starting point for any discussion of whether centralising or decentralising the provision of a public service such as health

care should be expected to improve outcomes. The decentralisation theorem developed by W. E. Oates (1972) holds that, absent economies of scale or uncorrected spillovers across jurisdictions, decentralised provision of a public good will always yield social welfare at least as high as centralised provision, because local governments can better match heterogeneous local preferences (Gadenne and Singhal 2013). A complementary strand, beginning with Musgrave (1959) and formalised in the tradition of Tiebout (1956), addresses the assignment of fiscal responsibility, arguing that efficiency losses are minimised when the level of government that provides a service also finances it through something approximating a benefits tax (Gadenne and Singhal 2013). A “second generation” of fiscal-federalism theory, associated with Besley and Coate (2003) and reviewed by W. Oates (2005), incorporates political-economy considerations — for instance, the possibility that locally elected representatives have conflicting preferences over the allocation of centrally pooled resources — while a separate line following Brennan and Buchanan (1980) frames decentralisation as a disciplining device against the expansionary tendencies of government, operating through inter-jurisdictional mobility or “yardstick competition” among comparable local governments (Gadenne and Singhal 2013).

Despite the World Bank’s promotion of decentralisation as a governance reform since the 1990s, Gadenne and Singhal (2013) show that developing countries remain considerably less fiscally decentralised — particularly on the revenue side — than developed countries today, and than the United States and United Kingdom were in the early twentieth century, largely because subnational governments in developing economies depend heavily on intergovernmental transfers rather than own-source revenue. The limited quasi-experimental evidence on the causal effects of decentralisation in developing countries is mixed: Faguet’s analysis of Bolivia’s 1994 reform finds that decentralisation increased local public investment in education and health and made it more responsive to local need (Faguet 2004); Kis-Katos and Sjahrir (2017) find comparable infrastructure gains from Indonesia’s 2001 decentralisation, concentrated in previously under-served districts; while cross-country studies of decentralisation’s effect on growth and other outcomes reach conflicting conclusions depending on specification and country sample (Gadenne and Singhal 2013).

More recent scholarship has turned to the reverse process — recentralisation — which several countries, including Mexico, have pursued in response to fiscal austerity, the operational demands of the COVID-19 pandemic, or broader political centralisation. A very recent study by Olvera et al. (2026) develops a context-sensitive theoretical framework specifically for the Mexican case, arguing that recentralisation involves a fundamental trade-off between capacity gains and proximity losses. On one side, concentrating financial, administrative, and human resources at the national level can improve service delivery by exploiting economies of scale, professionalising bureaucracies, and enabling more standardised provision — advantages that are particularly salient in a technically demanding sector such as health care, where procurement capacity and administrative coordination matter a great deal. On the other side, decentralised systems facilitate accountability and responsiveness by placing decision-making closer to citizens, allowing services to be tailored to local conditions and lowering the cost of monitoring provider performance; recentralisation risks eroding these proximity advantages by weakening information flows between citizens and distant central authorities, generating confusion over entitlements, and reducing the responsiveness of service provision to territorial heterogeneity (Olvera et al. 2026).

Whether the capacity gains or the proximity losses dominate is, in this framework,

contingent on political and institutional context rather than fixed by the reform's design alone. Olvera et al. (2026) emphasises vertical partisan alignment between the federal government and state governments as a key moderating factor: where the state and federal governments share the same political party, coordination costs are expected to be lower and subnational governments have stronger incentives to cooperate with, rather than resist, a centrally driven reform, potentially allowing national administrative capacity to complement rather than substitute for local capacity. Under partisan misalignment, by contrast, overlapping responsibilities between levels of government can produce resistance or strategic non-compliance by subnational actors, amplifying proximity losses even where formal administrative capacity has increased. This theoretical expectation is directly relevant to the Mexican case, since the eight states that declined to sign coordination agreements with INSABI, and later with IMSS-Bienestar, were disproportionately governed by opposition parties, suggesting that political alignment may help explain not only which states recentralised but also the outcomes observed where they did.

The empirical record on recentralisation in the health sector specifically is described by Olvera et al. (2026) as inconclusive. Malesky et al. (2014) find that Vietnam's fiscal, administrative, and political recentralisation was associated with improvements in public-health infrastructure and reductions in health-care fees, while the study of township recentralisation in Taiwan by Wang (2024) finds reductions in the provision of several public goods, including water coverage and social infrastructure. Within health care narrowly, Simonet (2022) associates recentralisation with reduced provider autonomy and weaker patient engagement, whereas other studies report increased utilisation of health facilities after recentralisation (Phommasack et al. 2005) or improved patient outcomes and reduced costs in other settings — a pattern of divergent findings that Olvera et al. (2026) attribute to differences in the institutional and political contexts in which recentralisation reforms have been implemented.

Complementing this economic and political-science literature, work in public administration on policy coordination offers a further lens through which to interpret the Mexican reform. Cejudo and Michel (2017) distinguishes coordination, coherence, and integration as related but analytically distinct solutions to the problem of fragmented government action, arguing that policy integration — the alignment of strategic and administrative decisions around a shared complex problem that exceeds the scope of any single agency's mandate — requires more than the mere transfer of formal authority from one level of government to another. Applied to the Mexican health reform, this distinction cautions against treating the transfer of administrative responsibility from states to INSABI, and subsequently to IMSS-Bienestar, as self-evidently equivalent to a reduction in fragmentation: whether the reform achieves integration in Cejudo and Michel's sense, rather than a mere relocation of coordination costs, depends on the extent to which the federal government develops the decision-making mechanisms, shared information systems, and administrative routines needed to genuinely unify — rather than simply centralise — the delivery of care to the population without contributory social security.

In sum, this literature yields no unambiguous prediction for the Mexican case. The classic fiscal-federalism tradition would predict welfare losses from centralising a heterogeneous public service such as primary health care, absent strong economies of scale; the capacity-gains literature emphasised by Olvera et al. (2026) suggests the opposite is plausible in a context of chronically weak subnational administrative capacity; and both traditions agree that the realised effect is likely to be conditional

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on contextual factors rather than uniform across the Mexican states that adopted the 2020 reform on different timelines and to different degrees. This theoretical ambiguity, combined with the staggered, state-level variation in adoption, motivates this research identification strategy.

## Chapter 2

# Methods

### 2.1 Data

The empirical analysis draws on a household-level panel constructed from five waves (2016, 2018, 2020, 2022, and 2024) of Mexico’s National Household Income and Expenditure Survey (Encuesta Nacional de Ingresos y Gastos de los Hogares, ENIGH) (INEGI 2024), merged with a state-year panel of health-system covariates. For each surveyed household, the panel includes quarterly out-of-pocket (OOP) health expenditure (`oop_expenditure`), decomposed into three mutually exclusive and exhaustive subcategories — medicines, hospitalisation (`hospital`), and ambulatory/outpatient care (`outpatient_care`) — which sum exactly to `oop_expenditure`, together with the household’s state of residence, survey year, an indicator for whether the household resides in a state classified as “treated”, and a set of household-level covariates: log household income, household head’s sex, age, and educational attainment, household size, the economic dependency ratio, and whether the household resides in a rural locality. With data taken from the General Directorate of Health Information, this study also controls for four supply-side indicators: the share of the state population living in poverty and the number of hospital beds, doctors, and outpatient consultations per 100,000 inhabitants. Households in the top 0.01% of the income distribution are trimmed throughout, since a small number of extreme earners otherwise dominate the Poisson likelihood used in the main specification (see Section 2.2). The resulting analytical sample used in the main regressions comprises 289,817 household-year observations, spanning all 32 states, of which 8 have never affiliated with INSABI/IMSS-Bienestar and thus constitute the control group, and 24 are classified as treated.

Table 2.1 summarises the two groups, pooling all five survey waves. Households in control states report higher mean OOP spending than those in treated states across all four spending measures — total, medicines, hospital, and outpatient/ambulatory care — and are drawn from a somewhat less severely uninsured, less impoverished population, even though supply-side conditions (hospital beds per 100,000 inhabitants) are broadly comparable between the two groups. Treated states, despite their higher poverty and uninsurance rates, record more doctors and outpatient consultations per 100,000 inhabitants than control states — a supply/need mismatch that is itself suggestive of why formal administrative capacity need not translate directly into lower out-of-pocket exposure. These level differences motivate the two-way fixed-effects design developed in Section 2.2, which nets out any time-invariant differences of this kind between treated and control states and instead identifies the reform’s effect from each group’s own change over time.

TABLE 2.1: Descriptive statistics by treatment status, pooled across all five survey waves (2016–2024). All means are weighted by the ENIGH household expansion factor. “Uninsured share of population” is the weighted share of each group’s own population classified as belonging to the population without contributory social security. State covariates are averaged across the state-year cells represented in the analytical sample.

|                                               | Treated  | Control  |
|-----------------------------------------------|----------|----------|
| Household-year observations                   | 210,342  | 79,475   |
| <b>Mean quarterly OOP expenditure (pesos)</b> |          |          |
| Total                                         | 1,229.60 | 1,457.10 |
| Medicines                                     | 275.01   | 278.82   |
| Hospital                                      | 166.63   | 288.53   |
| Outpatient/ambulatory care                    | 787.95   | 889.75   |
| Uninsured share of population (%)             | 71.3     | 59.5     |
| <b>State covariates</b>                       |          |          |
| Population in poverty (%)                     | 41.1     | 26.3     |
| Hospital beds per 100,000                     | 71.8     | 72.5     |
| Doctors per 100,000                           | 156.5    | 135.8    |
| Outpatient consultations per 100,000          | 60,381   | 46,253   |

Descriptively, OOP spending is heavily zero-inflated, which shapes the choice of estimator described below: 38.9% of households report no quarterly OOP health spending at all. This zero mass is far from uniform across subcategories. Hospitalisation has a 97.4% zero share; an expected quarter with no hospitalisation event is the norm, even though hospitalisation is comparatively costly when it occurs. While medicines and outpatient care have zero shares of 59.2% and 60.5%, respectively (see Figure A.1 in the Appendix). Baseline (pre-2020, control-group) mean quarterly spending also differs substantially by subcategory, at roughly 156 pesos for medicines, 147 pesos for hospital, and 776 pesos for outpatient care, out of a total mean of approximately 1,079 pesos — meaning outpatient/ambulatory care accounts for the large majority of total OOP spending in levels, independently of any treatment effect.

## 2.2 Identification strategy

The analysis exploits the staggered, state-level adoption of the 2020 (re)centralisation reform in a two-way fixed-effects difference-in-differences design. States and survey years enter as fixed effects, so that the coefficient of interest is the interaction term `treated_x_post`, comparing the change in outcomes between the pre- and post-reform periods across states that transferred administration of their health services to the federal government and the eight states that did not. The 8 never-treated states serve as the control group throughout.

Because ENIGH survey waves are biennial (2016, 2018, 2020, 2022, 2024) while most states’ actual transfer of authority occurred across 2020–2021, the single survey wave closest to the reform — 2020 — captures a mixture of already-transferred and not-yet-transferred treated states. The main specification defines `post = 1[year ≥ 2020]`, which places the entire adoption window at or after the first post-reform wave and is the specification reported as the headline result. Because this choice is not innocuous, three robustness checks vary the treatment-timing assumption: an earlier cutoff

(`post = 1[year = 2018]`), a later cutoff (`post = 1[year = 2022]`, which reclassifies 2020 itself as pre-reform), and a specification that drops the 2020 wave from the estimation sample entirely. A complementary event-study specification replaces the single `treated_x_post` interaction with separate `treated × year` indicators for 2016, 2018, 2022, and 2024 (2020 as the reference year), which allows the pre-reform years to be inspected directly for evidence of differential trends between treated and control states ahead of the reform (see Figure 3.1).

Given that 38.9% of the main outcome and up to 97.4% of one subcategory are exactly zero, the main model is estimated by survey-weighted Poisson pseudo-maximum-likelihood (PPML) on the raw peso outcome, rather than by ordinary least squares on a log-transformed outcome. This choice follows recent methodological guidance (Chen and Roth 2024; Bellemare and Wichman 2020) showing that the conventional  $\log(1 + y)$  transformation produces semi-elasticities that are not invariant to the added constant and can materially overstate treatment effects precisely in settings with a substantial zero mass, as confirmed for this specific application in the robustness comparison reported in Section 3.2 (see Figure 3.2). All models weight observations by the ENIGH household expansion factor and cluster standard errors by state (32 clusters: 24 treated, 8 control). Given the small number of control clusters, cluster-robust standard errors are treated here as a first-pass approximation, but further research should apply a wild cluster bootstrap or other small-cluster correction to confirm the robustness of inference.

## 2.3 Outcome variable

The primary outcome is household quarterly out-of-pocket health expenditure in constant pesos (`oop_expenditure`) - adjust for inflation using the Bank of Mexico's official consumer price index - estimated on its natural (untransformed) scale via PPML as described above. Three secondary outcomes decompose this total into its constituent subcategories—medicines, hospitalisation, and outpatient/ambulatory care—using the identical specification, to identify which component of spending drives any aggregate effect (see Figure 3.3). A binary indicator of whether a household reported any OOP spending at all (`any_oop`) is used as a complementary extensive-margin outcome, estimated via logit average marginal effects, to distinguish an effect operating through the probability of incurring any OOP expense from one operating through the amount spent conditional on spending something. A log-transformed version of the outcome,  $\log(1 + \text{oop\_expenditure})$ , is retained only for the robustness comparisons. All models control for the same set of household covariates (log income, household head's sex, age, and education, household size, the economic dependency ratio, and a rural-locality indicator) and state-level supply covariates (poverty share, and hospital beds, doctors, and outpatient consultations per 100,000 population), in addition to state and year fixed effects — the full set of estimated covariate effects is reported in Figure A.2 in the Appendix.

## Chapter 3

# Results, discussion and conclusion

### 3.1 Event study

The event-study specification, which estimates the treated–control difference separately for each survey year, does not show a clean, flat pre-reform baseline. Relative to 2020, the treated–control gap in log expected OOP spending was significantly more negative in both 2016 ( $-0.337$ ,  $p < 0.001$ ) and 2018 ( $-0.221$ ,  $p = 0.003$ ), continued narrowing through 2022 ( $-0.146$ ,  $p = 0.006$ ), and became statistically indistinguishable from the 2020 baseline by 2024 ( $-0.061$ ,  $p = 0.372$ ). The gap narrows essentially monotonically across the full 2016–2024 window rather than flattening or reversing after the reform, which is, if anything, harder to reconcile with the parallel-trends assumption than a flatten-then-widen pattern would be: the post-reform years continue closing the same gap that was already closing beforehand, rather than showing a break at 2020. Figure 3.1 plots both the raw parallel-trends comparison and the event-study coefficients side by side: panel A shows mean quarterly OOP spending converging between treated and control states across the sample period, and panel B shows the corresponding regression-adjusted gap narrowing steadily from 2016 through 2024, illustrating the pattern described above. This pattern is discussed further as a limitation in Section 3.4.

### 3.2 Main effect

The main two-way fixed-effects PPML specification estimates that IMSS-Bienestar/IN-SABI affiliation is associated with a 23.0% increase in expected household OOP health spending (coefficient = 0.207, SE = 0.063, 95% CI: +8.7% to +39.2%,  $p = 0.001$ ), holding household and state covariates and state/year fixed effects constant. Among the control variables, log household income (+140.3%,  $p < 0.001$ ) and the economic dependency ratio (+72.5%,  $p < 0.001$ ) are the strongest predictors of OOP spending, and household head’s age (+1.0%,  $p < 0.001$ ) and education (+3.5%,  $p < 0.001$ ) are also positive and significant; household head’s sex and household size are not statistically significant. Households in a rural locality report 31.3% higher expected OOP spending than otherwise-similar households in urban localities ( $p < 0.001$ ), the largest household-level effect after income and dependency ratio — plausibly reflecting a combination of greater reliance on paid private or informal providers where public facilities are sparser, and higher effective prices for care (including transport) outside urban centres. None of the four state-level supply covariates (poverty share, hospital beds, doctors, or outpatient consultations per 100,000 population) is a significant predictor of household

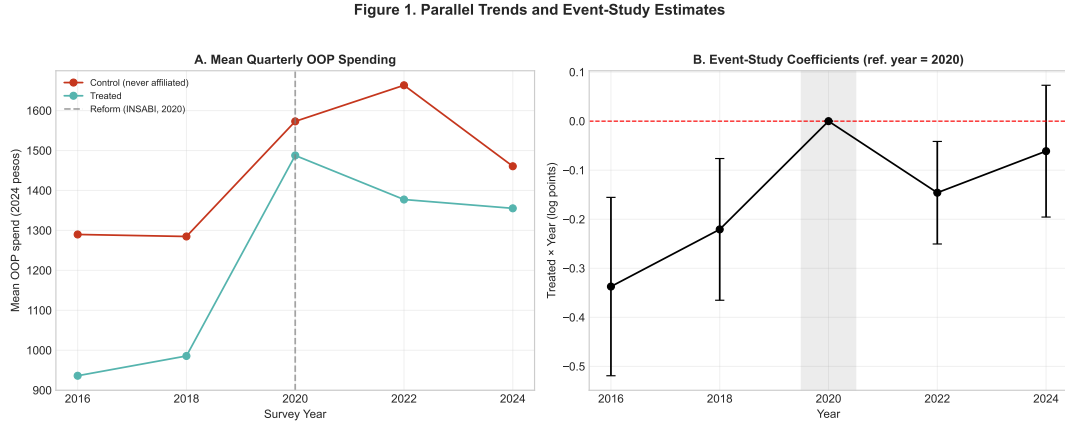


FIGURE 3.1: Parallel trends and event-study estimates. Panel A shows mean quarterly out-of-pocket spending by year and treatment status; panel B shows two-way fixed-effects event-study coefficients (treated  $\times$  year, PPML, relative to 2020) with 95% confidence intervals. The shaded band marks the 2020 survey wave, which straddles the actual 2020–2021 rollout of coordination agreements.

OOP spending once state and year fixed effects are included, which is expected given that these fixed effects absorb most of the cross-state, cross-time variation in supply conditions, leaving comparatively little within-state, within-year variation to identify their coefficients. The full set of covariate estimates is plotted in Figure A.2 and tabulated in full in Table A.1 in the Appendix.

### 3.3 Treatment timing and OOP decomposition

The magnitude and statistical significance of the main effect remain sensitive to how the ambiguous 2020 survey wave is classified. Moving the cutoff earlier, to 2018, strengthens the estimated effect to +25.7% ( $p = 0.003$ ). Moving it later, to 2022, reduces the estimate to +4.1% and renders it statistically indistinguishable from zero ( $p = 0.419$ ). Dropping the 2020 wave from the sample entirely yields an estimate in between, positive and now clearly significant (+17.0%,  $p = 0.027$ ). Two alternative estimators — log-OLS and survey-weighted WLS on  $\log(1 + \text{oop\_expenditure})$  — agree with the main model in direction and, under this control group, both clear conventional significance (+28.8%,  $p = 0.045$ ; and +50.6%,  $p = 0.027$ , respectively), though both still produce larger point estimates than PPML, consistent with the known tendency of the  $\log(1+y)$  transformation to overstate effects when a substantial zero mass is present. An extensive-margin logit specification, estimating the effect on the probability of reporting any OOP spending at all, finds a positive and, under this control group, statistically significant average marginal effect (+4.1 percentage points,  $p = 0.036$ ), suggesting that the effect identified in the main model operates both through drawing previously non-spending households into some OOP spending and through the amount spent by households that already incur some. Figure 3.2 summarises all seven specifications on a common percentage-change scale: every point estimate is positive, and six of seven are significant at conventional levels or better — the `post = 2022` cutoff is the only specification under which the effect is not detectable.

Applying the main specification separately to each of the three OOP subcategories

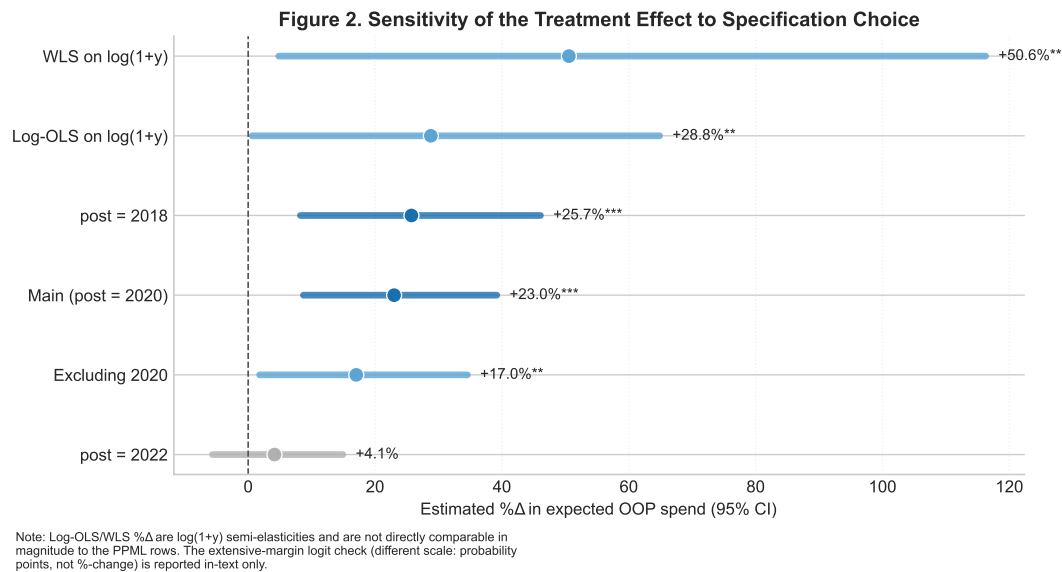


FIGURE 3.2: Sensitivity of the treatment effect to specification choice. Point estimates and 95% confidence intervals for `treated_x_post` across the main specification and six robustness checks. Log-OLS/WLS coefficients are  $\log(1+y)$  semi-elasticities and are not directly comparable in magnitude to the PPML rows; the extensive-margin logit check is on a different scale (probability points).

shows that the aggregate effect is not confined to a single category. Outpatient care rises by 16.6% (95% CI: +2.4% to +32.6%,  $p = 0.020$ ), hospital by 37.9% (95% CI: -1.4% to +93.0%,  $p = 0.061$ ), and medicines by 47.5% (95% CI: +9.7% to +98.2%,  $p = 0.010$ ): all three point estimates are positive, and medicines and outpatient care are each individually significant at the 5% level, with hospital significant at the 10% level. Outpatient care remains the most precisely estimated of the three in relative terms — its confidence interval is by far the tightest — and, in absolute peso terms, by far the largest component of total OOP spending, with a control-group pre-reform mean of approximately 875 pesos per quarter, roughly four times the baseline mean of either medicines (193 pesos) or hospital (220 pesos). Medicines has both the largest point estimate of the three and the strongest evidence against a null effect; hospital's confidence interval, by contrast, is by far the widest and only barely excludes zero, consistent with its extreme (97.4%) zero share leaving comparatively little quarter-to-quarter variation to estimate from. A simple approximation (baseline mean multiplied by estimated percentage change) attributes about 92 pesos to medicines, 83 pesos to hospital care, and 145 pesos to outpatient services, totaling roughly 320 pesos, somewhat above the approximately 300-peso increase indicated by refitting the total model on this same outcome (1,287 pesos times 23.3%); this gap reflects the fact that summing three separately-fit, nonlinear (log-link) Poisson means need not reconcile exactly with the total's own fit. Read together, the evidence points to an effect that is broadly distributed across all three categories of OOP spending rather than concentrated in any one of them, with outpatient/ambulatory care contributing the most in absolute pesos, medicines carrying the most statistically secure relative estimate of the three, and hospital contributing a comparably large but least precisely estimated share. Figure 3.3 shows the four estimates — total spending and the three subcategories — on the same axis.

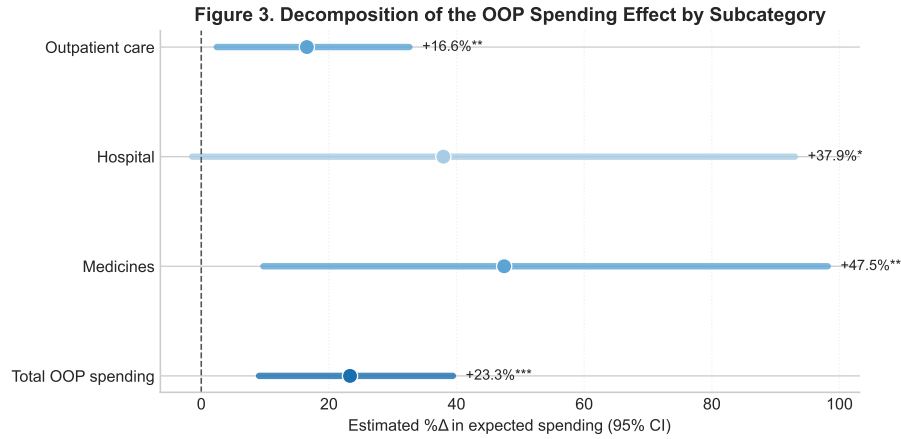


FIGURE 3.3: Decomposition of the out-of-pocket spending effect by subcategory. Point estimates and 95% confidence intervals for `treated_x_post` (PPML, main specification) applied separately to total spending and each of its three constituent subcategories.

### 3.4 Discussion

The central empirical finding — that affiliation with the federally centralised health institutions (INSABI, subsequently IMSS-Bienestar) is associated with an increase, rather than a decrease, in household out-of-pocket health spending — sits uncomfortably next to the reform’s stated objective, reiterated in the 2030 goals discussed in Section 1.3.3, of reducing precisely this form of financial exposure. It is, however, broadly consistent with the household-survey evidence reviewed in Section 1.4: Knaul et al. (2023) and Serván-Mori et al. (2023) both documented a marked deterioration in financial protection following the 2019–2020 transition from Seguro Popular to INSABI, including a near-doubling of excessive health expenditure and a sharp rise in average out-of-pocket spending per user between the 2018 and 2021 rounds of ENSANUT. The present results extend that evidence by benchmarking the effect against a genuine control group of states that never transferred administration of their health services, rather than relying solely on before–after comparisons for the treated population as a whole, and by decomposing it across spending categories rather than reporting only a single aggregate figure.

The wide-ranging impact across different spending subcategories can be understood within the theoretical framework. According to Olvera et al. (2026), “proximity losses”—which refer to the decrease in responsiveness, information exchange, and accountability of decentralized, locally managed systems—serve as the main way through which centralizing reforms might actually weaken service delivery, even when official administrative capacity is increasing. Since proximity losses tend to affect households’ trust in and access to the overall public system—not just a single point of contact—a general rise in out-of-pocket (OOP) spending can be explained by this mechanism, just as well as an increase limited to a specific category.

Outpatient care remains the level of care where continuity with a known local provider, familiarity with entitlements, and short travel distances matter most, and where Knaul et al. (2023) separately documented rising waiting times and increased reliance on private providers after 2018, which plausibly explains why its estimate is by far the largest in absolute peso terms and the tightest in relative (percentage) terms; medicines, in fact, returns the strongest statistical evidence of the three (the lowest

p-value and largest point estimate), consistent with routine medicine purchases being a similarly accessible and immediate substitution margin for households facing reduced confidence in, or availability of, public-facility care. Hospital shows a comparably large point estimate but, for the purely mechanical reason that hospitalisation events are rare (a 97.4% zero share), is estimated with markedly less precision in a panel of this size, not necessarily because the underlying effect is smaller or absent.

A plausible reading of the subcategory results, then, is that households facing confusion over entitlements or reduced confidence in the reorganised public network responded by paying out of pocket more broadly across the care they sought — whether in private facilities or as informal payments within the public system — rather than substituting specifically at the outpatient margin. This pattern also speaks directly to the fragmentation concerns raised in Section 1.2: rather than resolving the financial-protection failures long attributed to Mexico’s fragmented health architecture, the 2020 reform’s initial years appear, on this evidence, to have reproduced or even amplified them for the specific population whose institutional arrangement it was designed to change.

That said, the headline 23.0% estimate should not be reported as a settled or fully robust finding, for two distinct reasons. First, the event-study results in Section 3.2 (Figure 3.1) show a treated–control gap that was already narrowing before 2020 and continues narrowing on essentially the same trajectory afterward, which is difficult to reconcile with the parallel-trends assumption that underlies the difference-in-differences design; if treated and control states were on differentially evolving trajectories prior to the reform for reasons unrelated to it, the estimated `treated_x_post` coefficient will conflate the reform’s true effect with the continuation of that pre-existing trend, and some part of the estimated increase could reflect trend continuation rather than a reform-induced break. Second, and relatedly, the magnitude and significance of the main effect remain somewhat sensitive to how the 2020 ENIGH wave — the survey round that straddles the actual, staggered 2020–2021 rollout of coordination agreements — is classified: treating it as post-reform or excluding it altogether both yield significant positive effects, while treating it as pre-reform effectively eliminates the effect (Figure 3.2). Both patterns point to the same underlying difficulty: the reform’s real-world timing was staggered and gradual in a way that a biennial household survey and a single common cutoff date cannot cleanly capture. A further caveat concerns inference: with only 8 control-state clusters, the cluster-robust standard errors reported here should be treated as a first approximation, and a wild cluster bootstrap — appropriate for designs with few clusters — is recommended before these results are presented as final.

None of these caveats overturns the basic direction of the finding but they do mean the appropriate claim is a qualified one: OOP health spending appears to have risen following the transfer of health-service administration to the federal government, an effect visible across all three spending subcategories and most clearly detected when the 2020 survey wave is coded as post-reform, but this finding is not yet robust to how that one transition-period wave is treated, and the pre-2020 event-study pattern cautions against a purely causal reading without further work to address the apparent pre-trend.

### 3.5 Conclusion

This research set out to examine whether Mexico’s (re)centralisation of health services for the population without contributory social security — first through INSABI, and subsequently through the expanded IMSS-Bienestar — improved or worsened household financial protection, exploiting the staggered, state-level adoption of the 2020 reform against a control group of eight states that never transferred administration to the federal government. The central empirical result is that affiliation with the centralised institutions is associated with a statistically significant 23.0% increase in expected household out-of-pocket health spending (95% CI: +8.7% to +39.2%), an effect visible across outpatient care, hospitalisation, and medicines alike, largest in absolute peso terms for outpatient/ambulatory care and statistically strongest for medicines. This finding runs counter to the reform’s stated 2030 objective of reducing out-of-pocket exposure, and is consistent both with earlier survey-based evidence of deteriorating financial protection after 2018 and with the “proximity losses” mechanism identified in the theoretical literature on recentralisation, whereby weakened local responsiveness and confusion over entitlements can push households toward private, out-of-pocket alternatives even as nominal public coverage expands.

This research explicitly shows that its main result is fragile along a specific dimension: its magnitude and statistical significance depend somewhat on how the ambiguous 2020 survey wave is classified. Additionally, the event-study analysis raises concerns about potential different pre-reform trends between treated and control states, a concern this control group does not resolve. Instead of considering the +23.0% figure as a definitive causal estimate, the study recommends presenting it as the primary result alongside all robustness checks. The main qualitative conclusion — that centralization is broadly associated with increased OOP spending across categories, statistically strongest for medicines and largest in absolute terms for outpatient care, but not yet fully robust to how the transition year is handled or to the apparent pre-trend — is considered a more reliable headline at this stage. Future research should use a wild cluster bootstrap due to the limited number of control clusters, seek administrative or high-frequency data to pinpoint each state’s transfer date more accurately than a biennial survey, and examine the mechanisms behind this shift, such as substitution to private providers, non-adopting state governments’ compensatory actions, or declining confidence in reorganized public facilities that may explain why out-of-pocket costs across categories seem to be absorbing the burden of this institutional change.

This estimate should also be read as a snapshot of an early and still-incomplete transition rather than of the centralised model’s steady state. Only eight states remained outside IMSS-Bienestar by 2024, and the panel analysed here covers, at most, the first two to four years of a rollout that continues to unfold nationally. Under the capacity-gains/proximity-losses framework this research relies on, that timing matters: the proximity losses associated with disrupting locally embedded arrangements — confusion over entitlements, weakened information flows between citizens and a distant federal authority — are largely front-loaded into the transition itself, while the capacity gains that motivate recentralisation in the first place, namely economies of scale, professionalising bureaucracies, and enabling more standardised provision (Olvera et al. 2026), plausibly take longer to materialise, as they depend on procurement systems, supply chains, and administrative practice consolidating at the national level. As more states affiliate and that consolidation proceeds, the expected balance could shift, with capacity gains overtaking the accountability and

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responsiveness advantages of decentralised provision that this research finds households currently worse off without.

# References

- Asociación Mexicana de Instituciones de Seguros. 2026. “Centro Estadístico Del Sector Asegurador Mexicano.” In *Centro Estadístico*. <https://centroestadisticoamis.mx/>.
- Bellemare, Marc F., and Casey J. Wichman. 2020. “Elasticities and the Inverse Hyperbolic Sine Transformation.” *Oxford Bulletin of Economics and Statistics* 82 (1): 50–61. <https://doi.org/10.1111/obes.12325>.
- Besley, Timothy, and Stephen Coate. 2003. “Centralized Versus Decentralized Provision of Local Public Goods: A Political Economy Approach.” *Journal of Public Economics* 87 (12): 2611–37. <https://ideas.repec.org/a/eee/pubeco/v87y2003i12p2611-2637.html>.
- Bossert, Thomas, Nathan Blanchet, Suzanne Sheetz, Diana M. Pinto, Jonathan Cali, and Ricardo Perez-Cuevas. 2014. “Comparative Review of Health System Integration in Selected Countries in Latin America.” *IDB Publications*, ahead of print, January. <https://doi.org/10.18235/0009151>.
- Brennan, Geoffrey, and James Buchanan. 1980. *The Power to Tax*. Cambridge University Press. <https://EconPapers.repec.org/RePEc:cup:cbooks:9780521233293>.
- Cejudo, Guillermo M., and Cynthia L. Michel. 2017. “Addressing Fragmented Government Action: Coordination, Coherence, and Integration.” *Policy Sciences* 50 (4): 745–67. [https://ideas.repec.org/a/kap/policy/v50y2017i4d10.1007\\_s11077-017-9281-5.html](https://ideas.repec.org/a/kap/policy/v50y2017i4d10.1007_s11077-017-9281-5.html).
- Chen, Jiafeng, and Jonathan Roth. 2024. “Logs with Zeros? Some Problems and Solutions.” *The Quarterly Journal of Economics* 139 (2): 891–936. <https://doi.org/10.1093/qje/qjad054>.
- CIEP. 2021. *De Seguro Popular a INSABI: Mayor Población Con Menor Atención*. <https://ciep.mx/de-seguro-popular-a-insabi-mayor-poblacion-con-menor-atencion/>.
- Cortés-Adame, Luis Javier, and Octavio Gómez-Dantés. 2025. “The Termination of Seguro Popular: Impacts on the Care of High-Cost Diseases in the Uninsured Population in Mexico.” *Lancet Regional Health – Americas* 46 (April): 101078. [https://www.thelancet.com/journals/lanam/article/PIIS2667-193X\(25\)00088-2/fulltext](https://www.thelancet.com/journals/lanam/article/PIIS2667-193X(25)00088-2/fulltext).
- Diario Oficial de la Federación, DOF -. 2003. *DECRETO Por El Que Se Reforma y Adiciona La Ley General de Salud*. DOF - Diario Oficial de La Federación. [https://dof.gob.mx/nota\\_detalle.php?codigo=695626&fecha=15/05/2003#gsc.tab=0](https://dof.gob.mx/nota_detalle.php?codigo=695626&fecha=15/05/2003#gsc.tab=0).

- DOF. 2019. *DECRETO Por El Que Se Reforman, Adicionan y Derogan Diversas Disposiciones de La Ley General de Salud y de La Ley de Los Institutos Nacionales de Salud*. [https://dof.gob.mx/nota\\_detalle.php?codigo=5580430&fecha=29/11/2019#gsc.tab=0](https://dof.gob.mx/nota_detalle.php?codigo=5580430&fecha=29/11/2019#gsc.tab=0).
- DOF. 2022. *DECRETO Por El Que Se Crea El Organismo Público Descentralizado Denominado Servicios de Salud Del Instituto Mexicano Del Seguro Social Para El Bienestar (IMSS-BIENESTAR)*. [https://dof.gob.mx/nota\\_detalle.php?codigo=5663064&fecha=31/08/2022#gsc.tab=0](https://dof.gob.mx/nota_detalle.php?codigo=5663064&fecha=31/08/2022#gsc.tab=0).
- Faguet, Jean-Paul. 2004. “Does Decentralization Increase Government Responsiveness to Local Needs?: Evidence from Bolivia.” *Journal of Public Economics* 88 (3-4): 867–93. <https://ideas.repec.org//a/eee/pubeco/v88y2004i3-4p867-893.html>.
- Gadenne, Lucie, and Monica Singhal. 2013. *Decentralization in Developing Economies*. Working Paper. Working Paper Series. National Bureau of Economic Research. <https://doi.org/10.3386/w19402>.
- Garcia-Diaz, Rocío, Sandra G. Sosa-Rubí, Rafael Lozano, and Edson Serván-Mori. 2023. “Equity in Out-of-Pocket Health Expenditure: Evidence from a Health Insurance Program Reform in Mexico.” *Journal of Global Health* 13: 04134. <https://doi.org/10.7189/jogh.13.04134>.
- García-Díaz, Rocio. 2022. “Effective Access to Health Care in Mexico.” *BMC Health Services Research* 22 (1): 1027. <https://doi.org/10.1186/s12913-022-08417-0>.
- Gómez-Dantés, Octavio, Laura Flamand, Diego Cerecero-García, Mariana Morales-Vazquez, and Edson Serván-Mori. 2023. “Origin, Impacts, and Potential Solutions to the Fragmentation of the Mexican Health System: A Consultation with Key Actors.” *Health Research Policy and Systems* 21 (1): 80. <https://doi.org/10.1186/s12961-023-01025-2>.
- INEGI. 2024. *Encuesta Nacional de Ingresos y Gastos de Los Hogares (ENIGH) 2024*. Instituto Nacional de Estadística y Geografía(INEGI). <https://www.inegi.org.mx/programas/enigh/nc/2024/>.
- Kis-Katos, Krisztina, and Bambang Suharnoko Sjahrir. 2017. “The Impact of Fiscal and Political Decentralization on Local Public Investment in Indonesia.” *Journal of Comparative Economics* 45 (2): 344–65. <https://doi.org/10.1016/j.jce.2017.03.003>.
- Knaul, Felicia Marie, Hector Arreola-Ornelas, Michael Touchton, et al. 2023. “Setbacks in the Quest for Universal Health Coverage in Mexico: Polarised Politics, Policy Upheaval, and Pandemic Disruption.” *Lancet (London, England)* 402 (10403): 731–46. [https://doi.org/10.1016/S0140-6736\(23\)00777-8](https://doi.org/10.1016/S0140-6736(23)00777-8).
- Knaul, Felicia Marie, Eduardo González-Pier, Octavio Gómez-Dantés, et al. 2012. “The Quest for Universal Health Coverage: Achieving Social Protection for All in Mexico.” *Lancet (London, England)* 380 (9849): 1259–79. [https://doi.org/10.1016/S0140-6736\(12\)61068-X](https://doi.org/10.1016/S0140-6736(12)61068-X).
- Knaul, Felicia Marie, Rebeca Wong, Héctor Arreola-Ornelas, Oscar Méndez, and

- Network on Health Financing and Social Protection in Latin America and the Caribbean (LANET). 2011. "Household Catastrophic Health Expenditures: A Comparative Analysis of Twelve Latin American and Caribbean Countries." *Salud Publica De Mexico* 53 Suppl 2: s85–95.
- Malesky, Edmund J., Cuong Viet Nguyen, and Anh Tran. 2014. "The Impact of Recentralization on Public Services: A Difference-in-Differences Analysis of the Abolition of Elected Councils in Vietnam." *American Political Science Review* 108 (1): 144–68. <https://scholars.duke.edu/publication/1028906>.
- Musgrave, Richard A. 1959. *The Theory of Public Finance: A Study in Public Economy*. McGraw-Hill.
- Oates, W. E. 1972. *Fiscal Federalism*. Harcourt Brace Jovanovich.
- Oates, Wallace. 2005. "Toward A Second-Generation Theory of Fiscal Federalism." *International Tax and Public Finance* 12 (4): 349–73. <https://ideas.repec.org/a/kap/itaxpf/v12y2005i4p349-373.html>.
- Olvera, Johabed G., Guillermo M. Cejudo, and Julio Alberto Ramos Pastrana. 2026. *Re-Centralization and Healthcare Service Delivery: Capacity Gains, Proximity Losses, and Partisan Alignment in Mexico*. SSRN Scholarly Paper. Social Science Research Network. <https://doi.org/10.2139/ssrn.6337878>.
- Phommasack, Bounlay, Lathiphone Oula, Oukeo Khounthalivong, et al. 2005. "DECENTRALIZATION AND RECENTRALIZATION: EFFECTS ON THE HEALTH SYSTEMS IN LAO PDR." *SOUTHEAST ASIAN J TROP MED PUBLIC HEALTH*.
- Serván-Mori, Edson, Octavio Gómez-Dantés, David Contreras, et al. 2023. "Increase of Catastrophic and Impoverishing Health Expenditures in Mexico Associated to Policy Changes and the COVID-19 Pandemic." *Journal of Global Health* 13: 06044. <https://doi.org/10.7189/jogh.13.06044>.
- Simonet, Daniel. 2022. "Centralization of Health Policy Decisions, Conversion to Austerity, and Public Defiance in Contemporary France." *Journal of Public Affairs* 22 (4): e2634. <https://doi.org/10.1002/pa.2634>.
- Tiebout, Charles M. 1956. "A Pure Theory of Local Expenditures." *Journal of Political Economy* 64 (5): 416–24. <https://www.jstor.org/stable/1826343>.
- Wang, Hsu Yumin. 2024. "Centralization, Elite Capture, and Service Provision: Evidence From Taiwan." *Comparative Political Studies* 57 (1): 32–68. <https://doi.org/10.1177/00104140231169023>.
- World Bank. 2025. *National Health Compact - Republic of Mexico*. <https://thedocs.worldbank.org/en/doc/0273f33ab6ee48c5d842108b9b55c789-0140022025/related/National-Health-Compact-Mexico.pdf>.

## Appendix A

# Supplementary Figures and Tables

This appendix reports supplementary figures and tables referenced from the main text: the zero-inflated distribution of out-of-pocket spending that motivates the choice of PPML over a log transformation (Section 2.2), the full set of estimated covariate effects from the main model in both plotted (Figure A.2) and tabulated (Table A.1) form (Section 3.2), the underlying marginal relationships for income and treatment, event-study estimates run separately for each spending subcategory, and a level comparison of mean spending by subcategory and treatment status. All figures and tables use the identical model specification and data as the corresponding main-text results; the code that produces them is available in the [GitHub Repository](#) alongside the rest of the empirical analysis.

Table A.1 reports the same estimates underlying Figure A.2 in full numerical detail.

TABLE A.1: Full coefficient table for the main two-way fixed-effects PPML model (survey-weighted, cluster-robust SE by state;  $N = 289,817$ ; 32 state clusters — 24 treated, 8 control). Coefficients are on the log scale;  $\% \Delta = 100 \cdot (\exp(\text{coef}) - 1)$  is the implied percentage change in expected OOP spending for a one-unit change in the regressor, with `treated_x_post` giving the headline DiD effect. State and year fixed effects (31 and 4 dummy coefficients, respectively) are included in the model but omitted from this table for brevity.

| Variable                                                   | Coef.   | SE     | p-value       | $\% \Delta$ | 95% CI ( $\% \Delta$ ) |
|------------------------------------------------------------|---------|--------|---------------|-------------|------------------------|
| Constant                                                   | −3.3524 | 0.6870 | <0.001<br>*** | —           | —                      |
| Treatment $\times$ Post<br>( <code>treated_x_post</code> ) | 0.2070  | 0.0631 | 0.001 ***     | +23.0%      | +8.7% to<br>+39.2%     |
| Log household income                                       | 0.8768  | 0.0401 | <0.001<br>*** | +140.3%     | +122.2% to<br>+160.0%  |
| Household head’s sex                                       | −0.0409 | 0.0255 | 0.109         | −4.0%       | −8.7% to<br>+0.9%      |
| Household head’s age                                       | 0.0102  | 0.0018 | <0.001<br>*** | +1.0%       | +0.7% to<br>+1.4%      |
| Household head’s<br>education                              | 0.0347  | 0.0068 | <0.001<br>*** | +3.5%       | +2.2% to<br>+4.9%      |
| Household size                                             | −0.0199 | 0.0123 | 0.106         | −2.0%       | −4.3% to<br>+0.4%      |

| Variable                             | Coef.   | SE     | p-value       | % $\Delta$ | 95% CI (% $\Delta$ ) |
|--------------------------------------|---------|--------|---------------|------------|----------------------|
| Economic dependency ratio            | 0.5452  | 0.0454 | <0.001<br>*** | +72.5%     | +57.8% to +88.5%     |
| Rural locality                       | 0.2723  | 0.0596 | <0.001<br>*** | +31.3%     | +16.8% to +47.6%     |
| State poverty share                  | 0.0010  | 0.0053 | 0.854         | +0.1%      | −0.9% to +1.1%       |
| Hospital beds per 100,000            | −0.0035 | 0.0041 | 0.387         | −0.4%      | −1.1% to +0.4%       |
| Doctors per 100,000                  | 0.0025  | 0.0020 | 0.203         | +0.2%      | −0.1% to +0.6%       |
| Outpatient consultations per 100,000 | 0.0000  | 0.0000 | 0.934         | +0.0%      | −0.0% to +0.0%       |

Figure A1. Zero-Inflated Distributions of OOP Spending

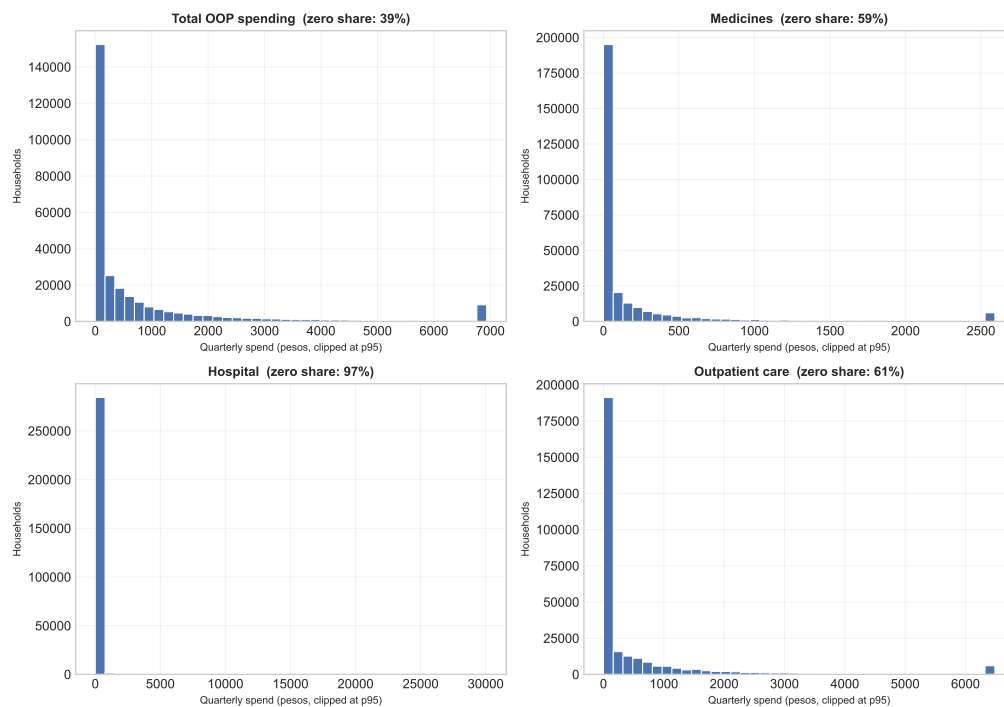


FIGURE A.1: Zero-inflated distributions of quarterly out-of-pocket spending, total and by subcategory. Hospitalisation spending in particular is dominated by zeros (97.4%), since a hospitalisation event in any given quarter is rare even though it is comparatively costly when it occurs.

Figure A2. Full Covariate Effects — Main PPML Model

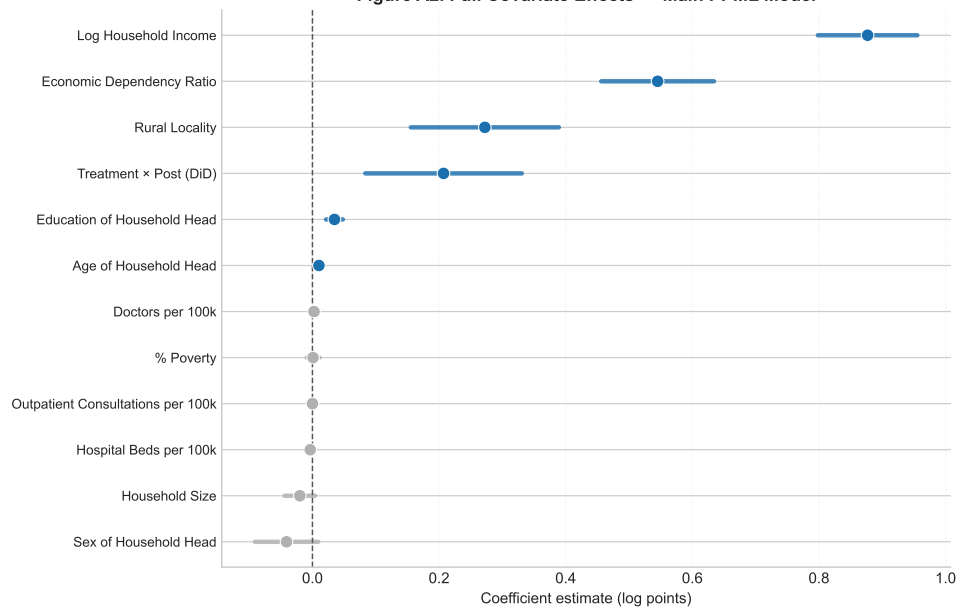


FIGURE A.2: Full covariate effects from the main survey-weighted PPML model. Point estimates and 95% confidence intervals for every household and state covariate, in addition to `treated_x_post`. State and year fixed effects are omitted.

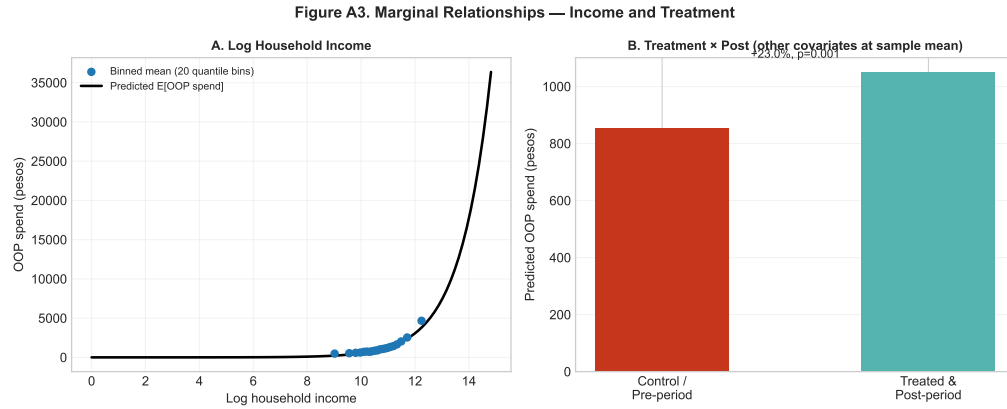


FIGURE A.3: Marginal relationships between predicted out-of-pocket spending and (A) log household income and (B) treatment status, holding other covariates at their sample means. Panel A uses binned means (20 quantile bins) rather than the raw household-level scatter, which at this sample size renders as an uninformative cloud.

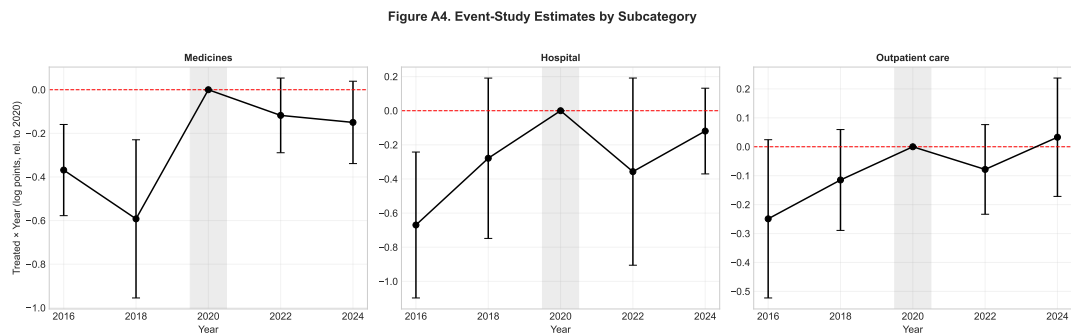


FIGURE A.4: Event-study estimates run separately for each of the three spending subcategories, using the identical specification as the main event study (Figure 3.1). The pre-2020 narrowing of the treated–control gap documented for total spending in Section 3.2 is not obviously concentrated in, or absent from, any single subcategory.

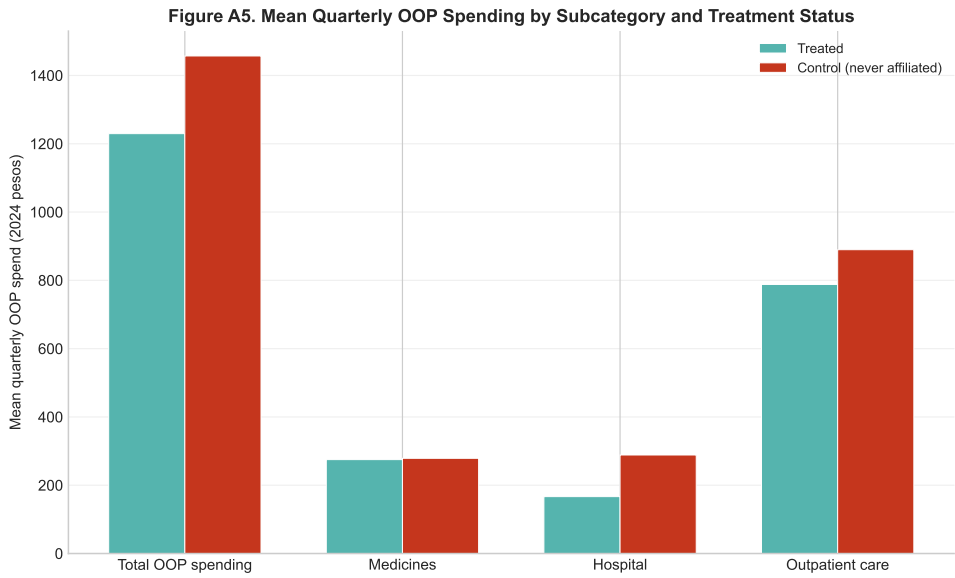


FIGURE A.5: Mean quarterly out-of-pocket spending by subcategory and treatment status, pooling all five survey waves. Complements the pre-reform baseline figures reported in Table 2.1 with the full-sample level comparison across treated and control states.